

RowIQ — Abishai Gosula

Pre-Work #7: MVP Finalization + Validation

UC Berkeley ENGIN 183C · Spring 2026 · Due April 2, 2026

Part 1 — MVP North Star

1.1 Reflection — Meta Goal

At the end of this semester, I want to walk into the April 23rd pitch competition and demo a live product that a Cal Rowing coach has actually used on the dock during practice. Not a mockup. Not a prototype with fake data. A real tool that replaced something they were doing manually and made their session noticeably easier. The moment that defines success for me is a coach using the boat assignment screen to build a lineup, publishing it, then timing their boats and recording splits during a live session — all without picking up a notebook or a separate stopwatch. That is the interaction that makes RowIQ undeniable. If I can show that moment to a VC or to Cal's Athletic Director on April 30th, with real session data behind it, I will be satisfied with the semester's output.

1.2 Overview — Core MVP Features

These three features are the minimum required to deliver core value to the coach. Everything else either builds on top of them or addresses secondary workflows.

Feature 1 — Boat Assignment + Lineup Publishing

- Drag-and-drop interface — coach assigns athletes to seats across 3 boats on one screen
- Visual rowing shell with port/starboard labels and 4-seat / 8-seat toggle
- Publish lineup — locks it, notifies athletes, transitions to live session
- Lineup history — every published lineup stored and viewable per athlete

Feature 2 — Strava-Style Live Session Stopwatch

- Per-boat independent stopwatches — large tap targets, one-handed dock use
- Split button burns a chapter marker into any active video recording at that exact timestamp
- Split log: split number, cumulative time, interval time, video marker indicator
- Swipeable boat cards — coach moves between boats mid-session without losing state
- All splits saved offline (IndexedDB) and synced to Supabase when back online

Feature 3 — Athlete Profile with Unified Data

- Per-athlete card: boat history, splits, wellness check-in trends, session timeline
- Bridge Athletics CSV import — athlete uploads gym export, coach sees load alongside water data
- Familiarity badges — 'Rowed together X times' computed from lineup history
- 'Last gym session' indicator on coach's roster view showing how fresh the Bridge data is

Why these three: The boat assignment tool is the feature no competitor offers and the one coaches feel most acutely every single practice. The stopwatch replaces the physical stopwatch and paper notebook in one screen — two tools eliminated at once. The athlete profile transforms RowIQ from a session tool into a platform — it is the data layer that makes the AI valuable later. Without all three, RowIQ is just another timer app. With all three, it is the first unified practice management system built for rowing.

1.3 Close But No Cigar — Pushed to Roadmap

The following features were considered for the MVP but deferred for the reasons listed below:

Google Drive OAuth + video sync: High technical complexity (OAuth, resumable upload, Wi-Fi detection). The core demo does not require it — splits and boat data are sufficient. Moving to Phase 2B after the pitch competition.

AI companion chatbot: Only valuable once real data exists from actual sessions. Building the AI before the data pipeline is complete produces meaningless responses. Scheduled for Phase 3A.

Wearable integration (Whoop, Garmin, Apple Health): Requires hardware partnership conversations and API approval. Adds significant complexity with marginal MVP value. Phase 3B.

Injury risk prediction: Requires multiple seasons of longitudinal data. Cannot build or validate this in three weeks. Phase 4.

Boat lineup optimizer: Coaches want control over lineup decisions. This becomes a recommendation layer once trust is established. Phase 4.

Drive spreadsheet viewer + AI queries: Depends on Drive integration and the AI layer, both of which are deferred. Phase 3.

1.4 Deeper — Feature Specifications

Feature 1 — Boat Assignment + Lineup Publishing

- Screen: PracticeSetupPage — opens from dashboard 'Start Practice' button
- 3 boats displayed simultaneously, horizontal scroll for additional boats (max 8)
- Each boat: SVG rowing shell, numbered seat circles (port = left, starboard = right), size toggle (4 or 8)
- Athlete roster below boats — drag onto seat to assign; drag back to unassign; drag onto occupied seat to swap
- Session name field at top; Add/Remove boat buttons
- Publish: confirmation modal → lineup locked in Supabase → athletes notified → transitions to Live Session
- Data: seat_assignments table (boat_id, athlete_id, seat_number, side, published_at)
- RLS: coaches see all lineups for their team; athletes see only lineups they are part of

Feature 2 — Strava-Style Live Session Stopwatch

- Screen: LiveSessionPage — full screen, no nav bar, exit only via 'End Session'
- Timer: `startTime = Date.now()` on Start; `elapsed = Date.now() - startTime` on each `requestAnimationFrame` tick
- Three bottom buttons: VIDEO (red pulsing when recording), SPLIT (logs split + chapter marker if recording), PAUSE
- Split log: scrollable — split number / cumulative time / interval time / video icon if marker exists
- Boat navigation: horizontal swipe or dot indicators; each boat has fully independent timer state
- Offline: splits saved to IndexedDB via `idb-keyval` on every tap; sync queue uploads to Supabase on reconnect
- Video: MediaRecorder API; video/mp4 on iOS, video/webm;codecs=vp9 on Android; blob stored locally
- Tag sheet on video stop: boat (pre-selected), athletes (multi-select from lineup), session ID + timestamps
- Post-session: auto-generates report — lineup, splits, voice notes, video — saved to `session_reports` table

Feature 3 — Athlete Profile with Unified Data

- Screen: AthleteProfilePage — accessible from coach roster and athlete's own Today screen
- Coach view: name, photo, boat history, current status, 7/30-day wellness trends, split trend chart
- Athlete view: next session, crew familiarity badges (rowed together N times), personal session timeline
- Bridge import: Profile > Connect Data > Bridge Athletics > upload CSV
- Parser: scan header row for patterns (date, exercise, sets, reps, weight) — no hardcoded column indices
- Preview screen: 'Found 47 sessions — import?' → saves to `bridge_sessions` and `bridge_exercises` tables
- Gym load score: $(total_sets \times avg_weight) / 100$, normalized 1-10, shown on coach's roster card
- Last gym indicator: 'Last gym: 2 days ago' on athlete card; turns yellow if more than 7 days old
- Duplicate detection: skip rows where date + exercise + sets + reps already exist

1.5 Reflection — Ambition Check

This scope is right on the money. Three features, each with a clear build path, zero dependency on external API partnerships, all buildable in Claude Code starting today. The boat assignment tool and stopwatch are already partially built in the current prototype — completing them to production quality is achievable in three weeks. The athlete profile with Bridge CSV import adds the data synthesis angle that makes RowIQ a platform rather than a utility. What makes this appropriately ambitious is that it is genuinely hard to build well — the offline stopwatch, the drag-and-drop on mobile, and the flexible CSV parser are all non-trivial — but none of them require external dependencies or partnership conversations. The features that were cut are genuinely Phase 2+ material. Removing them was not a cop-out — it was the right call for a three-week sprint.

Part 2 — Validation

2.1 Metric — How We Track Validation

Primary metric: Users — specifically active coaches and athletes inside the Cal Rowing pilot. Revenue is premature at this stage. Attention metrics (social impressions, waitlist) do not validate that the product actually works. What validates RowIQ at this stage is a coach opening the app before practice, building a lineup, running the session timer, and coming back the next day.

Validation targets:

Active coaches using the app: 2 coaches (Head + assistant, Cal Rowing) — By April 23rd

Athletes with active accounts: 30+ athletes across men's and women's programs — By April 23rd

Live sessions through RowIQ: 5+ real practice sessions logged — By April 23rd

Bridge CSV uploads: 10+ athletes upload their Bridge data — By April 30th

Verbal LOI from coaching staff: Coach AI formally confirms interest in continued use — By April 30th

Additional program waitlist: 3+ other collegiate programs request access — By May 8th

2.2 Execution — How We Get There

We can start immediately — no need to wait for the full MVP.

This week: Star, Rose, Lily, and Matt share the live prototype link (calrowing.vercel.app) with Cal Rowing coaches directly. We have team members on both squads — warm introduction, not cold outreach. Goal: get one coach to log in and assign a lineup before next class.

Before April 10th: Run the boat assignment tool during one actual practice. Coach uses it live on the dock. We collect feedback on what broke, what was confusing, and what they wished existed. This gives us a real usage session to reference in the April 23rd pitch.

Before April 23rd: 5 real sessions logged through RowIQ. At least 30 athlete accounts active. One coach gives us a direct quote to put on a slide — their words, not ours. Bridge CSV import tested with at least 10 athletes' real data.

Before April 30th: Verbal letter of intent from coaching staff confirming they want to continue using RowIQ after the semester. Begin outreach to 3-5 other D1 rowing programs: 'We're piloting at Cal — want early access?' This builds the waitlist for post-semester.

Ongoing: Weekly usage report shared internally: sessions logged, athletes active, features used most. Creates accountability and surfaces the data needed to show traction at the Collider Cup.

2.3 Reflection — Ambition Check

This is appropriately ambitious. The targets are specific and tied to real deadlines. Five sessions and thirty athletes by April 23rd is achievable because we have direct warm access to both Cal Rowing programs — no cold outreach, no app store submission, no marketing budget required. The LOI target is ambitious but credible: if a coach uses RowIQ for five real sessions and it saves them time, asking for a verbal commitment to continue is a natural conversation. The right metric here is honest validation — real coaches using a real tool in real sessions — not social impressions or waitlist signups from strangers.

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